



AI-Powered Mortgage Underwriting Engine

Agenda

01

Introduction

Overview of Awesome Tech, problem statement & importance, area of focus, chosen profile, why is now the correct time?

02

Approach & Methodology

Deep dive into our tech-stack and architecture, OCR capabilities, data validation pipelines, and the RAG frontend integration.

03

Conclusion & Next Steps

Analyzing the benefits, radical efficiency gains, unit economics, and our strategic roadmap for future implementation.

Introduction



Awesome Tech

We build future-ready software to make mortgage operations Easy, Safe and Secure



What's the problem?

Manual mortgage underwriting takes 3-5 days and costs \$2,400 per loan, with 70% of time spent on document review



Why it is important?

Delays causes borrower abandonment and cost the industry \$16B annually in manual processing

We are generating value by automating the Underwriting process:

Borrowers → Document collection → income verification → compliance check → risk assessment → approve/deny decision

Our Focus



Loan Type

Self-employed borrowers with inconsistent business income (e.g. C-Corps, S-Corps, partnerships)



Why

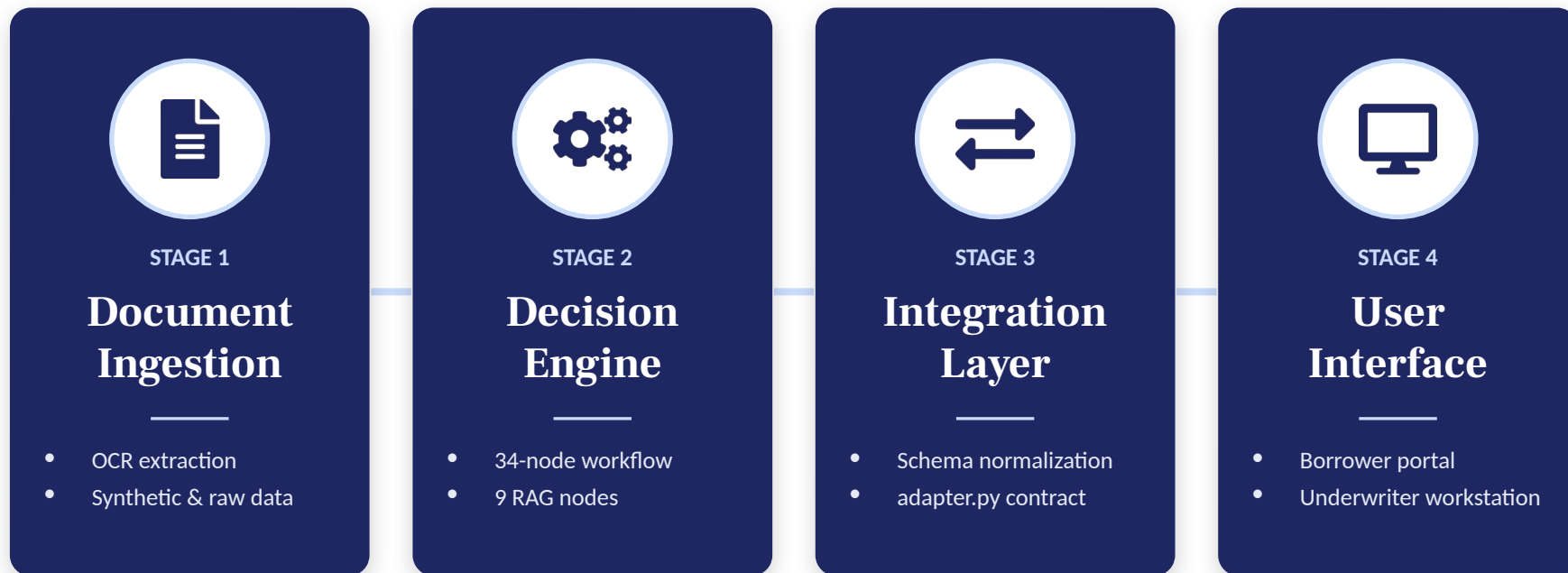
Fastest-growing borrower segment yet hardest to automate due to variable income, non-payroll deposits, and complex documentation



Breakthrough

2019: Standard OCR 60-70% accurate
2024: AWS Textract 95%+ extraction accuracy
2024: RAG + LLMs can cross-reference complex regulatory rules in real-time

The Solution: Intelligent Underwriting Engine



Amazon Textract (OCR)



Example from 1120-S Income Tax Return

Deductions	18	Employee benefit programs	18	
	19	Energy efficient commercial buildings deduction (attach Form 7205)	19	
	20	Other deductions (attach statement)	20	
	21	Total deductions. Add lines 7 through 20	21	15,000
	22	Ordinary business income (loss). Subtract line 21 from line 6	22	280,000

Doc. Type Query Based Extraction

OCR



1. What is the ordinary business income on line 22?
2. How much is the total assets in field F?
3. ...

```
"QUERY_RESULT": {  
  "Text": "280,000",  
  "Confidence": 98.7  
}
```

← Outputs field with
confidence score

Underwriting Decision Engine

Check
Eligibility



Document
Collection



Business
Type Income
Calculation



Trend
Analysis



Business
Health

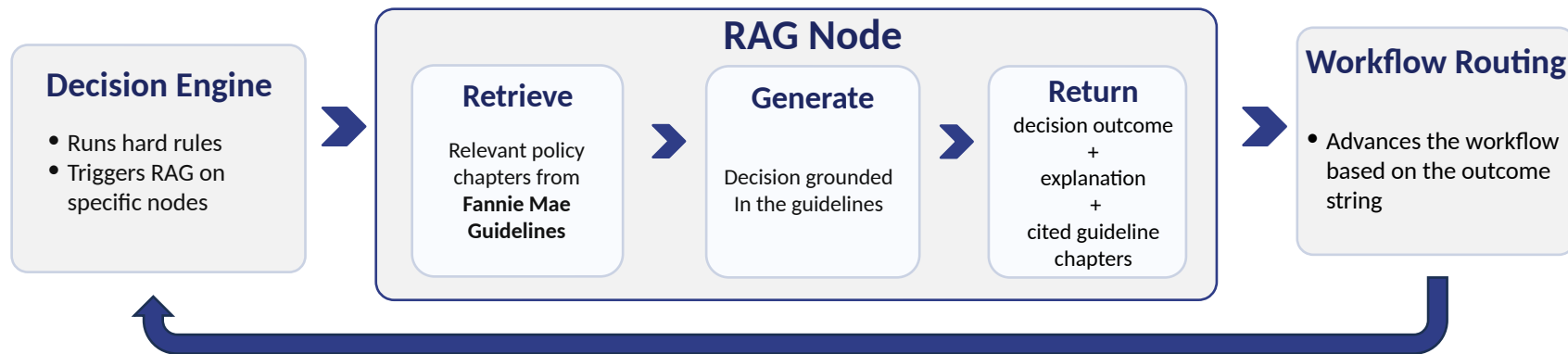


34 total nodes

RAG Integration

Policy-Grounded Underwriting Decisions

Where RAG plugs into our Workflow



Tech stack details

Retrieval layer

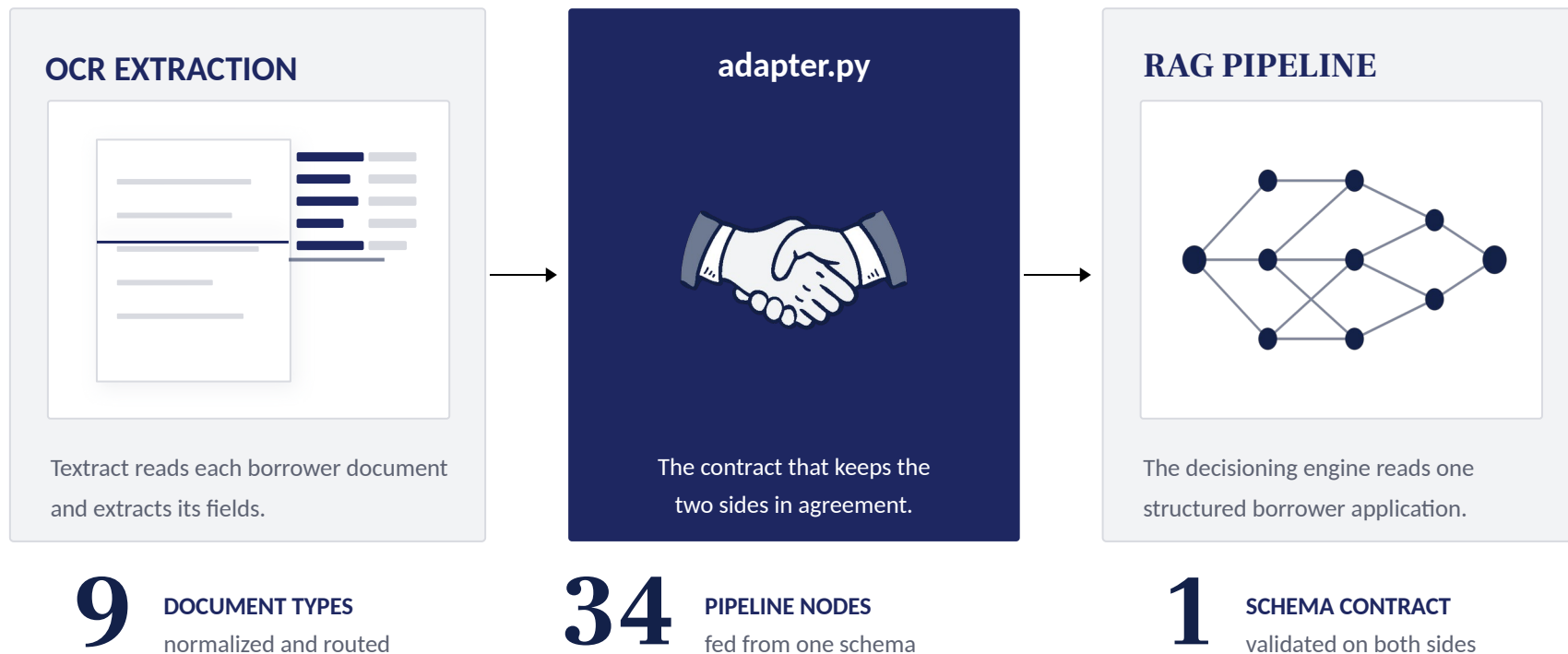
- **Chunks**: Fannie Mae chapter-based chunks
- **Embeddings**: BAAI/bge-small-en-v1.5
- **Vector search**: FAISS (cosine similarity)

Generation layer

- **Generation**: Groq API Llama 3.3 70B

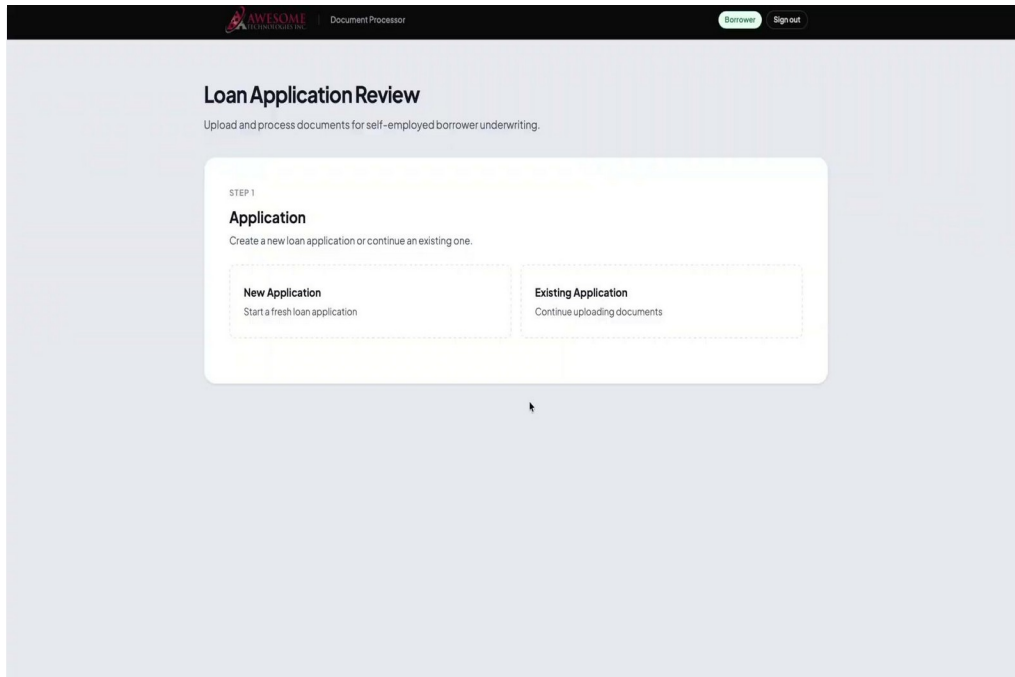
The Integration Layer

A single contract between how documents arrive and how decisions are made.



Built for the Borrower and Underwriter

Two Users, Two Interfaces, One Flow



Borrower Portal → Underwriter Workstation

1

SELF-EMPLOYED PORTAL

Borrower selects business structure and required docs generate

2

VERIFIED IN REAL TIME

Files flip from empty to amber to green as they upload

3

ONE-CLICK PIPELINE

Underwriter runs all 34 nodes from the workstation

4

DECISION DASHBOARD

Metrics, verification, and recommendation populate in real time

The Business Impact

Delivering measurable value across speed, accuracy, and regulatory compliance.

~95%

Extraction Accuracy

Eliminates Manual Touchpoints

Automatically structures complex financial data, removing human error.

3 Mins

Instant Eligibility

Reduces Borrower Abandonment

Converts a 3-day manual review into an immediate, automated decision.

100%

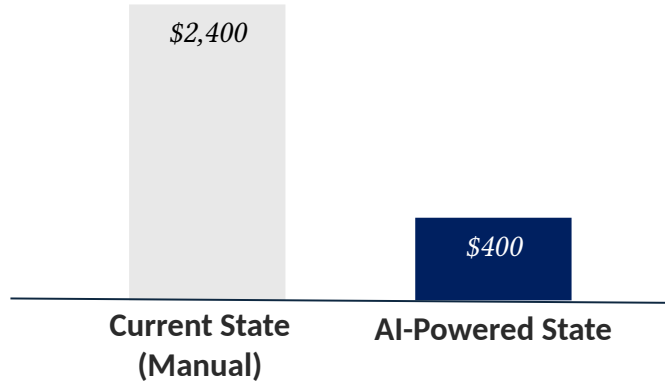
Risk Mitigation

Acts as a transparent co-pilot

Generates a strict 'Audit Trail' linking to Fannie Mae rules, ensuring zero hallucinations.

Unit Economics: Radical Efficiency

Total Cost per Loan Processed



Projected savings: \$2000 per loan

↓83% cost reduction

CURRENT STATE

67 hours

**Document Processing
Cost
\$1.50/doc**

**Manual Labor
~\$2,360**

Includes Underwriters, Processors, QA/Compliance Overhead

AI-POWERED STATE

3 min

**Document Processing
\$0.015/doc**

**First-Pass Labor
\$0**

*Powered by AWS Textract API & Human-in-loop validation
60% cases fully automated*

~30 docs per loan

Traction & Roadmap

This Isn't Just an Idea; We've Passed the Hardest Technical Hurdles

What We've Achieved



We aren't just building a tool, we're building the Operating System for modern automated lending

Acknowledgements

Faculty Advisors

Robert Hammond

Dan Mitchell

Sponsors

Noor Surani & Noor Punjwani

Project Team

BAX Team (Sydney Hinckley, Sheila Zhang, Tanushree Duggirala, Sanjana Ganesh)

University Support

UT Austin McCombs MS Staff

Thank You
Q&A